

Noise-Resistant Commonality and Individuality Label Learning for Multiview Multilabel Feature Selection Using Fuzzy Mutual Information

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Abstract—With the improvement of data collection capabilities and the increase in data volume, multiview multilabel feature selection has gained widespread attention. In multiview multilabel feature selection, it is inevitable to consider both the consistency and complementarity information between views. However, many existing methods tend to independently extract consensus and complementary information, and this fragmented approach often leads to ambiguity in the information partitioning, which in turn generates noise. Furthermore, existing methods do not consider the correlation between views and labels when estimating the effectiveness of different views, which can affect the accuracy of view weights. To address these issues, we propose a multiview multilabel feature selection method that integrates a noise-resistant strategy with fuzzy mutual information for jointly learning commonality and individuality label structures. Specifically, first, we learn the commonality label matrix using nonnegative matrix factorization to explore the consistency between views. This matrix not only captures shared patterns across views but also helps the model better fit the ground-truth labels. Second, we learn individuality label matrices to capture the complementary information of each view, and introduce view noise label matrices to mitigate the impact of inherent noise in the true labels and noise labels generated by partitioning. Third, we incorporate view weights based on fuzzy mutual information, which can accurately reflect the correlation between views and labels to enhance important views while reducing the contribution of noise. Finally, we design a sparse model-based multiview multilabel feature selection method and provide theoretical proof of its convergence. Extensive experiments on multiple benchmark datasets are conducted to demonstrate the effectiveness of our approach.

Index Terms—Commonality and individuality label learning, fuzzy mutual information, multiview multilabel feature selection, sparse model.

NOMENCLATURE

$\mathbf{X}^{(j)} \in \mathbb{R}^{n \times d(j)}$	Feature matrix in the j th view.
$\mathbf{Y} \in \mathbb{R}^{n \times c}$	Label matrix.
$\mathbf{W}^{(j)} \in \mathbb{R}^{d(j) \times c}$	Feature weight matrix in the j th view.
$\mathbf{W} \in \mathbb{R}^{d \times c}$	Feature weight matrix.
$\mathbf{C} \in \mathbb{R}^{n \times c}$	Commonality label matrix.
$\mathbf{Q}^{(j)} \in \mathbb{R}^{c \times d(j)}$	Coefficient matrix in the j th view.
$\mathbf{Y}_I^{(j)} \in \mathbb{R}^{n \times c}$	Individuality label matrix in the j th view.
$\mathbf{Y}_N^{(j)} \in \mathbb{R}^{n \times c}$	View noise label matrix in the j th view.
η_j	Weight of the j th view.

I. INTRODUCTION

WITH the explosive growth of data, extracting valuable information from vast datasets has become a significant challenge in the fields of machine learning and big data [1], [2], [3]. In this context, feature selection, as a concise and interpretable dimensionality reduction technique, can effectively address this challenge [4]. In particular, it has achieved significant success in single-view data [5], [6]. For example, Hu et al. [7] proposed an unsupervised feature selection method that leverages high-order similarity and sparse learning with an adaptive optimization strategy. Li et al. [8] proposed a supervised feature selection method combining trace ratio LDA with $l_{2,p}$ -norm regularization to select discriminative features for high-dimensional classification. However, applying these methods to multiview data scenarios introduces new difficulties and challenges. Unlike traditional single-view data, multiview data often comes from different perspectives or channels, and this diversity of data views increases the complexity of feature selection.

Existing multiview feature selection methods can be divided into three categories based on whether they utilize supervised information: unsupervised multiview feature selection [9], [10], [11], multiview single-label feature selection [12], [13], and multiview multilabel feature selection [14]. For example, in an unsupervised environment, Tang et al. [15] utilized the shared and discriminative information between different views through cross-view local structures and proposed an MV-UFS model. Yang et al. [16] proposed a tensor-based

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unsupervised feature selection framework to address data imbalance, incompleteness, and noise in multiview data. In a single-label environment, to handle large-scale multiview data, Lin et al. [17] proposed a multiview feature selection method based on shared submodels and local sparse regularization. In a multilabel environment, Hao et al. [18] proposed a latent representation method that learns a global view based on anchor points. Notably, among the three types of environments mentioned above, multiview multilabel feature selection is more complex due to its unique challenges. Specifically, in a multiview multilabel setting, data often originate from different perspectives, and data samples are typically described by multiple semantic labels [19], [20], [21]. For example, in social media analysis, a post can simultaneously contain text, images, and videos, and it may have multiple topic labels (such as “travel,” “food,” and “family”). The semantic information of these labels often needs to be comprehensively extracted from multiple views (such as text, images, and videos). This semantic complexity makes feature selection more challenging in multiview multilabel learning.

In recent years, several studies have focused on analyzing semantic complexity. Liu et al. [22] designed a label correlation matrix to fully utilize label information and mined the latent feature information and latent label information from multiview multilabel data. Liu et al. [23] constructed a common latent subspace across different views by utilizing the consistency relationship between labels and features. Although these two methods perform well in leveraging consensus information, they do not sufficiently explore the complementary information between views. Hao et al. [24] investigated view-specific label relationships and learned a specific label matrix for each view. Although this method explores view-specific information, it does not adequately consider the dominant role of view consistency relationships in multiview data. Moreover, existing methods often handle the consensus and complementary parts of views separately, but do not deal with the related labels accordingly, which may lead to noise due to uncertainty segmentation. Specifically, different views often contain individual information unique to each modality, which may result in view-specific labels. When such labels are absent or weakly supported in other views, they may act as noisy signals during joint learning, introducing uncertainty in the label assignment across views.

In addition, the importance of different views is often imbalanced. This imbalance may arise from differences in data quality, varying noise levels, or the characteristics of feature distributions between views. Wan et al. [25] adaptively learned effective view weights to balance views in an unsupervised multiview environment. However, this method is mainly designed for unsupervised scenarios and is not suitable for multiview multilabel environments, as the latter requires considering both label information and the relationship between view features. Hao et al. [26] proposed a view weight calculation method based on graph Laplacian theory in a multiview multilabel environment. This method uses the pairwise similarity between instances in the feature matrix and the inverse relationship of vector differences in the label matrix to derive view weights. However, this calculation

approach does not take into account the correlation between the views and the labels. Therefore, developing a more suitable and reasonable view weight calculation method for multiview multilabel environments remains a challenge.

To address the above challenges, by proposing a noise-resistant commonality and individuality label learning strategy and introducing fuzzy mutual information, this article constructs a multiview multilabel feature selection method. The method captures the shared patterns between multiple views by introducing a commonality label matrix and ensures that the model fits the true labels as much as possible while maintaining consistency across views by constraining the difference between the common label matrix and the true label matrix. To further explore the complementary information between views, we construct individuality label matrices and view noise label matrices, which represent the unique feature representations of each view and the influence of noise labels, respectively. By applying the l_1 -norm, we encourage sparsity in both the individuality label matrix and the view noise label matrix, ensuring the dominant role of the commonality label matrix. Furthermore, considering that different views may have varying effectiveness in describing the data, we compute the weight of each view through fuzzy mutual information to reasonably reflect the importance differences between views. The main contributions of this article are the following.

- 1) Our approach integrates the commonality label matrix and individuality label matrices to jointly capture both commonality and individuality information within multiview data, significantly improving the model’s ability to distinguish shared patterns and individual characteristics. Crucially, we construct view noise label matrices to effectively address noise caused by label segmentation and inherent label inconsistencies, thereby enhancing the model’s resilience to noisy data.
- 2) Based on fuzzy mutual information, we propose a view weight calculation method that effectively measures the correlation between each view and the labels, thereby reasonably allocating the importance weight of each view to enhance the contribution of important views while diminishing the effect of noise.
- 3) By integrating the denoised view label matrices with the view weights into a sparse model, we construct a provably convergent multiview multilabel feature selection method. This method allows for a more precise exploitation of interview individuality information, while effectively mitigating the impact of noise and view importance differences.

The remainder of this article has the following structure. Section II introduces the concepts of multiview multilabel data, fuzzy similarity relation, and fuzzy information measure. Section III presents our proposed multiview multilabel feature selection method. Section IV provides the solution of the proposed method and analyzes its convergence. The analysis of experimental results is presented in Section V. Finally, Section VI summarizes the research presented in this article.

II. PRELIMINARY KNOWLEDGE

A. Multiview Multilabel Data

Assume $MVMLD = \langle U, X, Y \rangle$ is a multiview multilabel dataset, where $U = \{u_1, u_2, \dots, u_n\}$ represents the sample set containing n samples, $X = \{X^{(j)}\}_{j=1}^v$ represents the collection of feature matrices from v views, with $X^{(j)} \in \mathbb{R}^{n \times d(j)}$ denoting the feature matrix of the j th view (containing n samples and $d(j)$ features), and $Y = \{l_1, l_2, \dots, l_c\}$ represents the nonempty label set with c class labels. For $\forall u_i \in U$, $u_i = \{x_i, y_i\}$, where $x_i = \{x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(v)}\}$ and $y_i = \{y_{i1}, y_{i2}, \dots, y_{ic}\}$ where $y_{ik} \in \{0, 1\}^{1 \times c}$. If the sample u_i is associated with the k th label, then $y_{ik} = 1$; otherwise, $y_{ik} = 0$.

B. Fuzzy Similarity Relation

Consider a nonempty finite set U , and let R be a fuzzy similarity relation [27] defined on U . The relation R assigns a value to each pair of elements in U , with the value lying between 0 and 1, where 1 represents complete similarity and 0 represents no similarity. For $\forall x_i, x_j \in U$, a fuzzy similarity relation is characterized by the following properties.

- 1) *Reflexivity*: $R(x_i, x_i) = 1$.
- 2) *Symmetry*: $R(x_i, x_j) = R(x_j, x_i)$.

A fuzzy similarity relation can be represented as a fuzzy similarity matrix [28] $FSM(R)$

$$FSM(R) = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{pmatrix} \quad (1)$$

where each entry r_{ij} represents the degree of similarity between the i th and j th samples in U . In practice, the value of r_{ij} can be defined using a Gaussian function: $r_{ij} = R(x_i, x_j) = \exp(-\|x_i - x_j\|^2 / (2 \cdot \sigma^2))$, and $r_{ij} \in [0, 1]$.

Definition 1 [29]: Suppose a fuzzy similarity relation R_B is induced by feature subsets B on U , a fuzzy similarity class $[x_i]_{R_B}$ is a fuzzy set combining membership degrees of all elements to x_i , expressed as: $[x_i]_{R_B} = r_{i1}/x_1 + r_{i2}/x_2 + \cdots + r_{in}/x_n$.

C. Fuzzy Information Measures

This section mainly introduces the methods of fuzzy information measurement, primarily covering the relevant knowledge of fuzzy information theory and fuzzy symmetric uncertainty.

Definition 2 [30]: For $\forall x_i \in U$, suppose the fuzzy similarity class $[x_i]_{R_B}$ is induced by B on U , the fuzzy information entropy of feature subsets B is defined as follows:

$$FIH(B) = -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{R_B}|}{n} \quad (2)$$

where $|[x_i]_{R_B}|$ represents the cardinality of the fuzzy similarity class $[x_i]_{R_B}$.

Definition 3 [31]: For $\forall x_i \in U$, suppose two fuzzy similarity class $[x_i]_{R_B}$, $[x_i]_{R_E}$ is induced by B and E on U ,

the fuzzy mutual information between feature subsets B and E is defined as follows:

$$FMI(B; E) = -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{n \cdot |[x_i]_{R_B} \cap [x_i]_{R_E}|}{|[x_i]_{R_B}| |[x_i]_{R_E}|}. \quad (3)$$

Fuzzy mutual information, a powerful metric for assessing the correlation between two variables, has strong interpretability and robustness, playing an important role in guiding filter-based feature selection [32], [33], [34].

Definition 4 [35]: For feature subsets B and E , the fuzzy symmetric uncertainty between feature subsets B and E is defined as follows:

$$FSU(B; E) = \frac{2 \cdot FMI(B; E)}{FIH(B) + FIH(E)}. \quad (4)$$

III. PROPOSED METHOD

This section mainly introduces the multiview multilabel feature selection method we proposed. The conceptual framework of our method is shown in Fig. 1.

A. Notations

In this article, we adopt the following notation: bold lowercase letters and bold uppercase letters denote vectors and matrices, respectively. Suppose $\mathbf{S} \in \mathbb{R}^{n \times m}$ denotes a matrix, $\mathbf{S}_{ij}$ represents the element located in the i th row and j th column of matrix $\mathbf{S}$, $\mathbf{S}_i$ denotes the i th row of matrix $\mathbf{S}$, and $\mathbf{S}_j$ denotes the j th column of matrix $\mathbf{S}$. The transpose and trace of matrix $\mathbf{S}$ are denoted by $\mathbf{S}^T$ and $\text{Tr}(\mathbf{S})$, respectively. The Frobenius norm, $l_{2,1}$ -norm and l_1 -norm of matrix $\mathbf{S}$ are represented as $\|\mathbf{S}\|_F = \left(\sum_{i=1}^n \sum_{j=1}^m \mathbf{S}_{ij}^2 \right)^{1/2}$, $\|\mathbf{S}\|_{2,1} = \sum_{i=1}^n \|\mathbf{S}_i\|_2 = \sum_{i=1}^n \left(\sum_{j=1}^m \mathbf{S}_{ij}^2 \right)^{1/2}$ and $\|\mathbf{S}\|_1 = \sum_{i=1}^n \sum_{j=1}^m |\mathbf{S}_{ij}|$. α, β, γ , and δ are the weighting parameters in the objective function that control the importance of each term. The notations utilized throughout this article are presented in Nomenclature.

B. Problem Formulation

In multiview multilabel feature selection, the core challenge lies in how to jointly select discriminative features across different views. Generally, the multiview multilabel feature selection problem can be formulated as the following optimization problem:

$$\min_{\mathbf{W}^{(j)}} \sum_{j=1}^v \left\| \mathbf{X}^{(j)} \mathbf{W}^{(j)} - \mathbf{Y} \right\|_F^2 + \Psi(\mathbf{W}) \quad (5)$$

where $\mathbf{X}^{(j)} \in \mathbb{R}^{n \times d(j)}$ and $\mathbf{W}^{(j)} \in \mathbb{R}^{d(j) \times c}$ represent the feature matrix and feature weight matrix of the j th view, respectively, $\mathbf{Y} \in \mathbb{R}^{n \times c}$ denotes the label matrix, $\mathbf{W} \in \mathbb{R}^{d \times c}$ represents the joint feature weight matrix of all views, and $\Psi(\bullet)$ is a regularization term used to control the model complexity.

Exploring the consistency and complementarity between views is crucial for improving model performance. To extract consistency information between views, we introduce the nonnegative matrix factorization (NMF) method, which learns a commonality label matrix ($\mathbf{C} \geq 0$) and coefficient matrix ($\mathbf{Q}^{(j)} \geq 0$), to capture shared patterns across multiple views.

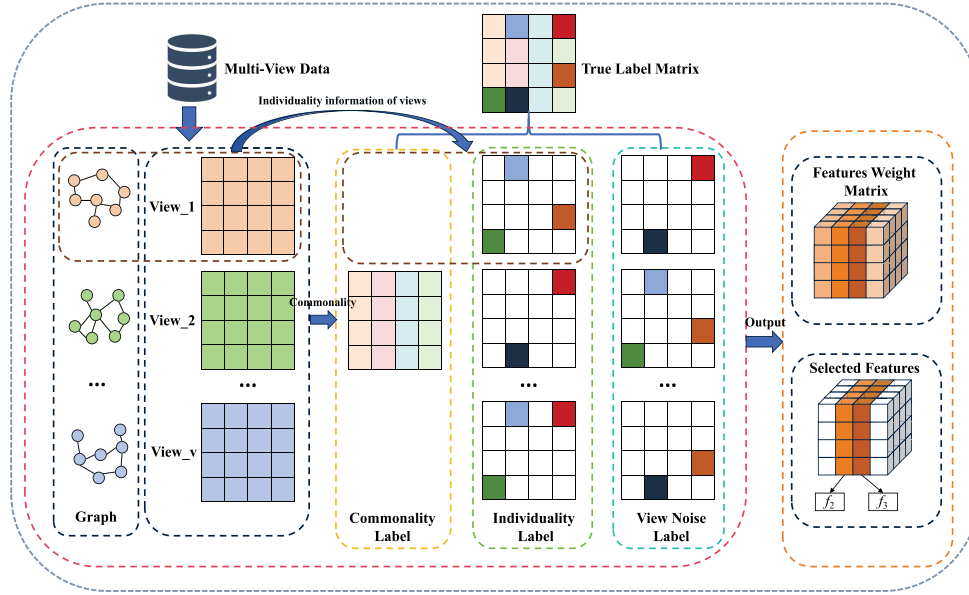


Fig. 1. Overview framework of our proposed NCILFS method. The true label matrix is decomposed into three components: commonality label, individuality label, and view noise label. The commonality label matrix reflects the consistency information between views, the individuality label matrices capture the complementary information between views, and the view noise label matrices are used to remove noise labels that affect model performance.

Considering that consistency between views is commonly observed in multiview data, and it reflects the common information in feature representations across different views, this commonality is often strongly correlated with the true labels of the data. We further introduce a constraint to limit the difference between the matrix $\mathbf{C}$ and the true label matrix $\mathbf{Y}$, in order to enhance the model's ability to fit the true labels while maintaining view consistency. Additionally, to select discriminative features, we apply the $l_{2,1}$ -norm to $\mathbf{W}$ to enforce row sparsity. Therefore, based on (5), the objective function can be rewritten as follows:

$$\min_{\mathbf{W}^{(j)}, \mathbf{Q}^{(j)}, \mathbf{C}} \sum_{j=1}^v \|\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \mathbf{C}\|_F^2 + \alpha \|\mathbf{Y} - \mathbf{C}\|_F^2 + \beta \sum_{j=1}^v \|\mathbf{X}^{(j)} - \mathbf{C} \mathbf{Q}^{(j)}\|_F^2 + \gamma \|\mathbf{W}\|_{2,1} \quad (6)$$

where $\mathbf{C} \in \mathbb{R}^{n \times c}$ is the commonality label matrix, and $\mathbf{Q}^{(j)} \in \mathbb{R}^{c \times d(j)}$ represents the coefficient matrix of the j th view.

In addition to considering the consistency between views, the different information between views (i.e., the complementary information between views) is also crucial. The complementarity between views reflects the unique feature representation of each view, which significantly contributes to the overall model performance. To capture the specific differential information between views, we introduce an individuality label matrix $\mathbf{Y}_I^{(j)}$, where $\mathbf{Y}_I^{(j)} \in \mathbb{R}^{n \times c}$ represents the individuality label matrix of the j th view. Since this individual information is exclusive, certain individual labels may act as noise labels for other views. In addition, the true labels themselves may also contain noise, which can negatively impact model fitting. To address this, we introduce a view noise label matrix $\mathbf{Y}_N^{(j)}$ to remove the influence of these noise labels, where $\mathbf{Y}_N^{(j)} \in \mathbb{R}^{n \times c}$ represents the noise label

matrix of the j th view. Thus, the model can focus on view-complementary information while avoiding interference from noise. Therefore, the true label matrix $\mathbf{Y}$ can be composed of the commonality label matrix $\mathbf{C}$, the individuality label matrix $\mathbf{Y}_I^{(j)}$, and the view noise label matrix $\mathbf{Y}_N^{(j)}$. To ensure that the relationships among these three components are reasonable and their differences are controllable, we add a constraint term in the objective function to limit the deviation between them. Consequently, the objective function can be rewritten based on (6) as follows:

$$\min_{\mathbf{W}^{(j)}, \mathbf{Q}^{(j)}, \mathbf{C}, \mathbf{Y}_I^{(j)}, \mathbf{Y}_N^{(j)}} \sum_{j=1}^v \|\mathbf{X}^{(j)} \mathbf{W}^{(j)} - (\mathbf{C} + \mathbf{Y}_I^{(j)})\|_F^2 + \alpha \sum_{j=1}^v \|\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)}\|_F^2 + \beta \sum_{j=1}^v \|\mathbf{X}^{(j)} - \mathbf{C} \mathbf{Q}^{(j)}\|_F^2 + \gamma \|\mathbf{W}\|_{2,1} \quad (7)$$

where $\tilde{\mathbf{Y}}^{(j)} = \mathbf{C} + \mathbf{Y}_I^{(j)} + \mathbf{Y}_N^{(j)}$.

Although each view contains its unique information, these views essentially represent different feature expressions of the same set of data. Therefore, from the perspective of data consistency, the commonality label matrix $\mathbf{C}$ should be the main component of the labels, while the individuality label matrix $\mathbf{Y}_I^{(j)}$ and the view noise label matrix $\mathbf{Y}_N^{(j)}$ should be sparse. To achieve this, we impose an l_1 -norm on $\mathbf{Y}_I^{(j)}$ and $\mathbf{Y}_N^{(j)}$ to encourage their sparsity. Moreover, the importance of different views in describing the data may not be the same. The more important a view is, the more view-complementary information it should provide (i.e., the information in $\mathbf{Y}_I^{(j)}$), and the less noise information it should have (i.e., the information in $\mathbf{Y}_N^{(j)}$). Therefore, to reasonably account for the importance differences between views, we further introduce view weights

into the objective function to improve overall performance. Based on these considerations, the objective function can be modified on the basis of (7) as follows:

$$\begin{aligned} \min_{\mathbf{W}^{(j)}, \mathbf{Q}^{(j)}, \mathbf{C}, \mathbf{Y}_I^{(j)}, \mathbf{Y}_N^{(j)}} & \sum_{j=1}^v \left\| \mathbf{X}^{(j)} \mathbf{W}^{(j)} - (\mathbf{C} + \mathbf{Y}_I^{(j)}) \right\|_F^2 \\ & + \alpha \sum_{j=1}^v \left\| \mathbf{Y} - \tilde{\mathbf{Y}}^{(j)} \right\|_F^2 \\ & + \beta \sum_{j=1}^v \left\| \mathbf{X}^{(j)} - \mathbf{C} \mathbf{Q}^{(j)} \right\|_F^2 + \gamma \|\mathbf{W}\|_{2,1} \\ & + \sum_{j=1}^v \left[(1 - \eta_j) \left\| \mathbf{Y}_I^{(j)} \right\|_1 + \eta_j \left\| \mathbf{Y}_N^{(j)} \right\|_1 \right] \end{aligned} \quad (8)$$

where η_j represents the weight of the j th view. To calculate the view weights, we use fuzzy mutual information to measure the correlation between each view and the labels. Fuzzy mutual information is an information-theoretic metric that can effectively capture the correlation between view features and labels, especially in situations where the feature distribution is complex in multiview data. Specifically, the calculation method for view weights is as follows:

Definition 5: Given a multiview multilabel dataset $\text{MVMLD} = \langle U, X \cup Y \rangle$, for the j th view feature set $X^{(j)} \subseteq X$, the view fuzzy symmetric uncertainty between $X^{(j)}$ and the label set Y is defined as

$$\begin{aligned} \text{VFSU}(X^{(j)}; Y) &= \sum_{f_w \in X^{(j)}} \text{FSU}(f_w; Y) \\ &= \sum_{f_w \in X^{(j)}} \sum_{l_i \in Y} \text{FSU}(f_w; l_i). \end{aligned} \quad (9)$$

Based on Definition 5, the weight of the j th view is designed as

$$\eta_j = \frac{\text{AVFSU}(X^{(j)}; Y)}{\sum_{j=1}^v \text{AVFSU}(X^{(j)}; Y)} \quad (10)$$

where

$$\text{AVFSU}(X^{(j)}; Y) = \frac{\text{VFSU}(X^{(j)}; Y)}{|X^{(j)}|} \quad (11)$$

where $|X^{(j)}|$ represents the number of features in $X^{(j)}$.

Motivated by the observation that “similar features share similar labels,” we believe that the local structure of the data should exhibit a certain degree of continuity in the label space. Specifically, in multiview learning, the local structure of the global view matrix should be consistent with that of the commonality label matrix $\mathbf{C}$. This is because the global view matrix weakens the differences between views, and its local structure can capture the common patterns of the data in the label space. Moreover, for each view, its feature matrix is not only related to the commonality label matrix $\mathbf{C}$ but also closely tied to its unique individuality label matrix $\mathbf{Y}_I^{(j)}$. Therefore, the local structure of the feature matrix of a single view should also align with the label matrix jointly formed by $\mathbf{C}$ and $\mathbf{Y}_I^{(j)}$. This consistency constraint ensures that the model can effectively leverage the complementary information unique to each view while mining the consistency across views. Based on the above analysis, we propose a regularization constraint to

strengthen the local structural consistency between the feature matrix and the label matrix

$$\min_{\mathbf{C}, \mathbf{Y}_I^{(k)}} \text{Tr}(\mathbf{C}^T \mathbf{L}_S \mathbf{C}) + \sum_{k=1}^v \text{Tr}(\tilde{\mathbf{C}}^{(k)T} \mathbf{L}_{S^{(k)}} \tilde{\mathbf{C}}^{(k)}) \quad (12)$$

where $\tilde{\mathbf{C}}^{(k)} = \mathbf{C} + \mathbf{Y}_I^{(k)}$, $\mathbf{L}_S = \mathbf{D} - \mathbf{S}$, $\mathbf{L}_{S^{(k)}} = \mathbf{D}^{(k)} - \mathbf{S}^{(k)}$. Here, $\mathbf{S} \in \mathbb{R}^{n \times n}$ is the global affinity matrix and $\mathbf{S}^{(k)} \in \mathbb{R}^{n \times n}$ is the affinity matrix of the k th view. $\mathbf{D} \in \mathbb{R}^{n \times n}$ and $\mathbf{D}^{(k)} \in \mathbb{R}^{n \times n}$ are diagonal matrices, with $\mathbf{D}_{ii} = \sum_{j=1}^n \mathbf{S}_{ij}$ and $\mathbf{D}_{ii}^{(k)} = \sum_{j=1}^n \mathbf{S}_{ij}^{(k)}$. $\mathbf{S}_{ij}$ and $\mathbf{S}_{ij}^{(k)}$ are defined as

$$\mathbf{S}_{ij} = \begin{cases} e^{-\frac{\|x_i - x_j\|^2}{\sigma^2}}, & x_i \in N_p(x_j) \text{ or } x_j \in N_p(x_i) \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

$$\mathbf{S}_{ij}^{(k)} = \begin{cases} e^{-\frac{\|x_i^{(k)} - x_j^{(k)}\|^2}{\sigma^2}}, & x_i^{(k)} \in N_p(x_j^{(k)}) \text{ or } x_j^{(k)} \in N_p(x_i^{(k)}) \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

where $N_p(x)$ represents the set of p -nearest neighbors of instance x , $x_i^{(k)}$ represents the i th sample in the k th view, and σ is a parameter that controls the width of the heat kernel.

By combining (8) and (12), the final objective function, including the constraints, is obtained as follows:

$$\begin{aligned} \min_{\mathbf{W}^{(j)}, \mathbf{Q}^{(j)}, \mathbf{C}, \mathbf{Y}_I^{(j)}, \mathbf{Y}_N^{(j)}} & \sum_{j=1}^v \left\| \mathbf{X}^{(j)} \mathbf{W}^{(j)} - \tilde{\mathbf{C}}^{(j)} \right\|_F^2 + \alpha \sum_{j=1}^v \left\| \mathbf{Y} - \tilde{\mathbf{Y}}^{(j)} \right\|_F^2 \\ & + \beta \sum_{j=1}^v \left\| \mathbf{X}^{(j)} - \mathbf{C} \mathbf{Q}^{(j)} \right\|_F^2 + \gamma \|\mathbf{W}\|_{2,1} \\ & + \sum_{j=1}^v \left[(1 - \eta_j) \left\| \mathbf{Y}_I^{(j)} \right\|_1 + \eta_j \left\| \mathbf{Y}_N^{(j)} \right\|_1 \right] \\ & + \delta \left[\text{Tr}(\mathbf{C}^T \mathbf{L}_S \mathbf{C}) + \sum_{k=1}^v \text{Tr}(\tilde{\mathbf{C}}^{(k)T} \mathbf{L}_{S^{(k)}} \tilde{\mathbf{C}}^{(k)}) \right] \\ \text{s.t. } & \mathbf{W}^{(j)}, \mathbf{Q}^{(j)}, \mathbf{C}, \mathbf{Y}_I^{(j)}, \mathbf{Y}_N^{(j)} \geq 0. \end{aligned} \quad (15)$$

IV. OPTIMIZATION

A. Solution

In (15), we need to solve for multiple variables. However, due to the presence of the $l_{2,1}$ -norm and l_1 -norm in the objective function, these norms render the objective function nonsmooth. To address this, we apply relaxation to the relevant terms [36] in (15): $\|\mathbf{W}\|_{2,1} = 2\text{Tr}(\mathbf{W}^T \tilde{\mathbf{D}} \mathbf{W})$, $\|\mathbf{Y}_I^{(j)}\|_1 = 2\text{Tr}(\mathbf{Y}_I^{(j)T} \mathbf{A}^{(j)} \mathbf{Y}_I^{(j)})$ and $\|\mathbf{Y}_N^{(j)}\|_1 = 2\text{Tr}(\mathbf{Y}_N^{(j)T} \mathbf{P}^{(j)} \mathbf{Y}_N^{(j)})$, where the diagonal matrix $\tilde{\mathbf{D}}_{ii} = 1/(2\|\mathbf{W}_i\|_2)$, $\mathbf{A}^{(j)} = 1/(2|\mathbf{Y}_I^{(j)}|)$ and $\mathbf{P}^{(j)} = 1/(2|\mathbf{Y}_N^{(j)}|)$.

Since the objective function includes nonnegativity constraints, we introduce Lagrange multipliers Λ , Γ , Φ , Ψ , and Ω to assist in the solution process. Subsequently, we adopt an iterative optimization approach to progressively solve for the values of each variable. Finally, the objective function can be rewritten in the following form:

$$\Theta = \sum_{j=1}^v \text{Tr} \left[(\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \tilde{\mathbf{C}}^{(j)})^T (\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \tilde{\mathbf{C}}^{(j)}) \right]$$

$$\begin{aligned}
& + \alpha \sum_{j=1}^v \text{Tr} \left[(\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)})^T (\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)}) \right] \\
& + \beta \sum_{j=1}^v \text{Tr} \left[(\mathbf{X}^{(j)} - \mathbf{CQ}^{(j)})^T (\mathbf{X}^{(j)} - \mathbf{CQ}^{(j)}) \right] \\
& + \delta \left[\text{Tr} (\mathbf{C}^T \mathbf{L}_S \mathbf{C}) + \sum_{k=1}^v \text{Tr} (\tilde{\mathbf{C}}^{(k)T} \mathbf{L}_{S^{(k)}} \tilde{\mathbf{C}}^{(k)}) \right] \\
& + 2 \sum_{j=1}^v \left[(1 - \eta_j) \text{Tr} \left(\mathbf{Y}_I^{(j)T} \mathbf{A}^{(j)} \mathbf{Y}_I^{(j)} \right) \right. \\
& \quad \left. + \eta_j \text{Tr} \left(\mathbf{Y}_N^{(j)T} \mathbf{P}^{(j)} \mathbf{Y}_N^{(j)} \right) \right] \\
& + 2\gamma \text{Tr} (\mathbf{W}^T \tilde{\mathbf{D}} \mathbf{W}) - \text{Tr} (\Lambda \mathbf{W}^{(j)T}) - \text{Tr} (\Gamma \mathbf{Q}^{(j)T}) \\
& - \text{Tr} (\Phi \mathbf{C}^T) - \text{Tr} (\Psi \mathbf{Y}_I^{(j)T}) - \text{Tr} (\Omega \mathbf{Y}_N^{(j)T}). \quad (16)
\end{aligned}$$

Then, by taking the derivatives of the objective function with respect to $\mathbf{W}^{(j)}$, $\mathbf{Q}^{(j)}$, $\mathbf{C}$, $\mathbf{Y}_I^{(j)}$, and $\mathbf{Y}_N^{(j)}$, we obtain

$$\frac{\partial \Theta}{\partial \mathbf{W}^{(j)}} = 2\mathbf{X}^{(j)T} (\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \mathbf{C} - \mathbf{Y}_I^{(j)}) + 4\gamma \tilde{\mathbf{D}}^{(j)} \mathbf{W}^{(j)} - \Lambda \quad (17)$$

$$\frac{\partial \Theta}{\partial \mathbf{Q}^{(j)}} = 2\beta \mathbf{C}^T \mathbf{CQ}^{(j)} - 2\beta \mathbf{C}^T \mathbf{X}^{(j)} - \Gamma \quad (18)$$

$$\begin{aligned}
\frac{\partial \Theta}{\partial \mathbf{C}} &= -2 \sum_{j=1}^v (\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \tilde{\mathbf{C}}^{(j)}) - 2\alpha \sum_{j=1}^v (\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)}) \\
&\quad - 2\beta \sum_{j=1}^v [(\mathbf{X}^{(j)} - \mathbf{CQ}^{(j)}) \mathbf{Q}^{(j)T}] + 2\delta \mathbf{L}_S \mathbf{C} \\
&\quad + 2\delta \sum_{j=1}^v \mathbf{L}_{S^{(k)}} \tilde{\mathbf{C}}^{(j)} - \Phi \quad (19)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \Theta}{\partial \mathbf{Y}_I^{(j)}} &= -2 (\mathbf{X}^{(j)} \mathbf{W}^{(j)} - \tilde{\mathbf{C}}^{(j)}) - 2\alpha \sum_{j=1}^v (\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)}) \\
&\quad + 2\delta \mathbf{L}_{S^{(k)}} \tilde{\mathbf{C}}^{(j)} + 4(1 - \eta_j) \mathbf{A}^{(j)} \mathbf{Y}_I^{(j)} - \Psi \quad (20)
\end{aligned}$$

$$\frac{\partial \Theta}{\partial \mathbf{Y}_N^{(j)}} = -2\alpha \sum_{j=1}^v (\mathbf{Y} - \tilde{\mathbf{Y}}^{(j)}) + 4\eta_j \mathbf{P}^{(j)} \mathbf{Y}_N^{(j)} - \Omega. \quad (21)$$

Let $\mathbf{M}^{(j)} = (\alpha + \delta \mathbf{D}^{(j)} + 1)(\mathbf{C} + \mathbf{Y}_I^{(j)}) + \alpha \mathbf{Y}_N^{(j)}$ and $\mathbf{H}^{(j)} = \mathbf{X}^{(j)} \mathbf{W}^{(j)} + \alpha \mathbf{Y} + \delta \mathbf{S}^{(j)}(\mathbf{C} + \mathbf{Y}_I^{(j)})$. By applying the KKT conditions, the following results are obtained:

$$\mathbf{W}^{(j)t+1} \leftarrow \mathbf{W}^{(j)t} \circ \frac{\mathbf{X}^{(j)T} (\mathbf{C} + \mathbf{Y}_I^{(j)})}{\mathbf{X}^{(j)T} \mathbf{X}^{(j)} \mathbf{W}^{(j)} + 2\gamma \tilde{\mathbf{D}}^{(j)} \mathbf{W}^{(j)}} \quad (22)$$

$$\mathbf{Q}^{(j)t+1} \leftarrow \mathbf{Q}^{(j)t} \circ \frac{\mathbf{C}^T \mathbf{X}^{(j)}}{\mathbf{C}^T \mathbf{CQ}^{(j)}} \quad (23)$$

$$\mathbf{C}^{t+1} \leftarrow \mathbf{C}^t \circ \frac{\sum_{j=1}^v (\mathbf{H}^{(j)} + \beta \mathbf{X}^{(j)} \mathbf{Q}^{(j)T}) + \delta \mathbf{SC}}{\sum_{j=1}^v (\mathbf{M}^{(j)} + \beta \mathbf{CQ}^{(j)} \mathbf{Q}^{(j)T}) + \delta \mathbf{DC}} \quad (24)$$

$$\mathbf{Y}_I^{(j)t+1} \leftarrow \mathbf{Y}_I^{(j)t} \circ \frac{\mathbf{H}^{(j)} + 2\eta_j \mathbf{A}^{(j)} \mathbf{Y}_I^{(j)}}{\mathbf{M}^{(j)} + 2\mathbf{A}^{(j)} \mathbf{Y}_I^{(j)}} \quad (25)$$

$$\mathbf{Y}_N^{(j)t+1} \leftarrow \mathbf{Y}_N^{(j)t} \circ \frac{\alpha \mathbf{Y}}{\alpha (\mathbf{C} + \mathbf{Y}_I^{(j)} + \mathbf{Y}_N^{(j)}) + 2\eta_j \mathbf{P}^{(j)} \mathbf{Y}_N^{(j)}} \quad (26)$$

where $\circ$ denotes the Hadamard product.

Algorithm 1 NCILFS Algorithm

Input: Feature matrices $\mathbf{X} = \{\mathbf{X}^{(j)}\}_{j=1}^v$; label matrix $\mathbf{Y}$; parameters α, β, γ , and δ .

Output: Set of selected features.

- 1: Calculate view weight η_j for each view using (10);
 - 2: Randomly initialize $\mathbf{W}^{(j)}$, $\mathbf{Q}^{(j)}$, $\mathbf{C}$, $\mathbf{Y}_I^{(j)}$, and $\mathbf{Y}_N^{(j)}$;
 - 3: **repeat**
 - 4: Update $\mathbf{W}^{(j)}$ using (22);
 - 5: Update $\mathbf{Q}^{(j)}$ using (23);
 - 6: Update $\mathbf{C}$ using (24);
 - 7: Update $\mathbf{Y}_I^{(j)}$ using (25);
 - 8: Update $\mathbf{Y}_N^{(j)}$ using (26);
 - 9: **until** convergence
 - 10: **return** $\mathbf{W}$.
-

The pseudocode of the proposed method is presented in Algorithm 1. There are five variables to be solved during the optimization process. To simplify, we suppose that the number of features in each view is d . When updating $\mathbf{W}^{(j)}$, $\mathbf{Q}^{(j)}$, $\mathbf{Y}_I^{(j)}$ and $\mathbf{Y}_N^{(j)}$, their time complexities are $O(nd^2c)$, $O(ndc^2)$, $O(n^2c)$, and $O(n^2)$, respectively. When updating $\mathbf{C}$, the time complexity is $O(n^2c)$ if $n \geq dc$, otherwise it is $O(ndc^2)$. Therefore, the time complexity of Algorithm 1 is $O(n^2c + nd^2c)$, since the number of features in a view is generally larger than the number of labels.

B. Convergence Analysis

This section covers a proof of the convergence of the proposed algorithm. For convenience, we denote $\mathbf{W}^{(j)}$, $\mathbf{X}^{(j)}$, $\mathbf{Y}_I^{(j)}$, $\tilde{\mathbf{D}}^{(j)}$, and $d(j)$ as $\mathbf{W}$, $\mathbf{X}$, $\mathbf{Y}_I$, $\tilde{\mathbf{D}}$, and d . The detailed process is as follows.

Definition 6: When $\mathbb{F}(\mathbf{W}, \mathbf{W}^t)$ adheres to the constraints that $\mathbb{F}(\mathbf{W}, \mathbf{W}) = \Theta(\mathbf{W})$ and $\mathbb{F}(\mathbf{W}, \mathbf{W}^t) \geq \Theta(\mathbf{W})$, it is referred to as an auxiliary function of $\Theta(\mathbf{W})$.

Lemma 1: The iterative procedure ensuring that $\Theta(\mathbf{W})$ does not increase can be formulated

$$\mathbf{W}^{t+1} = \arg \min_{\mathbf{W}} \mathbb{F}(\mathbf{W}, \mathbf{W}^t). \quad (27)$$

Proof: $\Theta(\mathbf{W}^{t+1}) \leq \mathbb{F}(\mathbf{W}^{t+1}, \mathbf{W}^t) \leq \mathbb{F}(\mathbf{W}^t, \mathbf{W}^t) = \Theta(\mathbf{W}^t)$.

The function $\Theta(\mathbf{W}_{ij})$ can be expanded into a Taylor series as shown below

$$\begin{aligned}
\Theta(\mathbf{W}_{ij}) &= \Theta(\mathbf{W}_{ij}^t) + \Theta'(\mathbf{W}_{ij}) (\mathbf{W}_{ij} - \mathbf{W}_{ij}^t) \\
&\quad + \frac{\Theta''(\mathbf{W}_{ij})}{2} (\mathbf{W}_{ij} - \mathbf{W}_{ij}^t)^2 \quad (28)
\end{aligned}$$

where

$$\Theta'(\mathbf{W}_{ij}) = (2\mathbf{X}^T (\mathbf{XW}_{ij} - \mathbf{C} - \mathbf{Y}_I) + 4\gamma \tilde{\mathbf{D}} \mathbf{W}_{ij})_{ij} \quad (29)$$

$$\Theta''(\mathbf{W}_{ij}) = (2\mathbf{X}^T \mathbf{X} + 4\gamma \tilde{\mathbf{D}})_{ii}. \quad (30)$$

Lemma 2: The following expression defines the auxiliary function $\mathbb{F}(\mathbf{W}_{ij}, \mathbf{W}_{ij}^t)$ of $\Theta(\mathbf{W}_{ij})$:

$$\begin{aligned}
\mathbb{F}(\mathbf{W}_{ij}, \mathbf{W}_{ij}^t) &= \Theta(\mathbf{W}_{ij}^t) + \Theta'(\mathbf{W}_{ij}) (\mathbf{W}_{ij} - \mathbf{W}_{ij}^t) \\
&\quad + \Delta(\mathbf{W}_{ij}^t) (\mathbf{W}_{ij} - \mathbf{W}_{ij}^t)^2 \quad (31)
\end{aligned}$$

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Dataset	Emotions	Yeast	Voc07	Mirflickr	Iaprtc12	Scene
Samples(n)	593	2417	3817	4053	4999	4400
Features(d)	72	103	712	712	1312	634
Labels(c)	6	14	20	38	260	33
view1	Rhythmic(8)	GE(79)	DH(100)	DH(100)	DH(100)	CH(64)
view2	Timbre(64)	PP(24)	GIST(512)	GIST(512)	DHV3H1(300)	CM(225)
view3			HH(100)	HH(100)	GIST(512)	CORR(144)
view4					HHV3H1(300)	EDH(73)
view5					HH(100)	WT(128)

where

$$\Delta(W_{ij}^t) = \frac{(X^T X W + 2\gamma \tilde{D} W)_{ij}}{W_{ij}^t}. \quad (32)$$

Proof: From (31), it is readily apparent that

$$\begin{aligned} (X^T X W + 2\gamma \tilde{D} W)_{ij} &= \sum_{h=1}^d (X^T X + 2\gamma \tilde{D})_{ih} W_{hj}^t \\ &\geq (X^T X + 2\gamma \tilde{D})_{ii} W_{ij}^t \end{aligned} \quad (33)$$

so $\Delta(W_{ij}^t) \geq (\Theta''(W_{ij}))/2$, then we can know $\mathbb{F}(W_{ij}, W_{ij}^t) \geq \Theta(W_{ij})$, and $\mathbb{F}(W_{ij}, W_{ij}^t) = \Theta(W_{ij})$ when $W_{ij} = W_{ij}^t$. Therefore, Lemma 2 is true.

We can establish the following iterative update rule for W based on Lemmas 1 and 2

$$\begin{aligned} W_{ij}^{t+1} &= W_{ij}^t - \frac{\Theta'(W_{ij}^t)}{2\Delta(W_{ij}^t)} \\ &= W_{ij}^t \frac{(X^T (C + Y_I))_{ij}}{(X^T X W + 2\gamma \tilde{D} W)_{ij}}. \end{aligned} \quad (34)$$

Similarly, the update rules for $Q^{(j)}$, C , $Y_I^{(j)}$ and $Y_N^{(j)}$ can be obtained, as indicated in (23)–(26). Consequently, when $W^{(j)}$, $Q^{(j)}$, C , $Y_I^{(j)}$ and $Y_N^{(j)}$ are updated according to Algorithm 1, the value of Θ does not increase.

V. EXPERIMENTAL ANALYSIS

A. Experimental Settings

The experiments are performed on six multiview multilabel datasets [18], [37], [38], with detailed information about each dataset presented in Table I. These datasets cover different application domains, as follows: yeast and emotions belong to the biological and music domains, respectively, and each has two views. Mirflickr, Voc07, Scene, and Iaprtc12 belong to the computer vision domain, with Mirflickr and Voc07 having three views, and Scene and Iaprtc12 having five views.

To highlight the superiority of the NCILFS algorithm, we compare it with nine advanced algorithms: 1) SSFS [39]; 2) GRRO [40]; 3) MDFS [41]; 4) LRDG [42]; 5) SRFS [43]; 6) MSFS [44]; 7) EF2FS [26]; 8) I2VSLC [24]; and 9) GRAFS [18]. The first five algorithms are classical single-view multilabel feature selection algorithms, while the last four are feature selection algorithms specifically designed for multiview multilabel learning. For single-view methods, all views are concatenated into a single feature space before feature selection. The parameter settings for

all comparison algorithms are configured according to the recommendations in the corresponding literature to ensure a fair comparison. Specifically, for SSFS and SRFS, the regularization parameter α, β , and γ are searched in the range $[0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 1]$. For GRRO, the parameter α_1 and β are searched in the range $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For MDFS and LRDG, the parameter α, β , and γ are searched in the range $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For MSFS, the parameter α, β , γ , and λ are searched in the range $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For EF2FS, I2VSLC, and GRAFS, the parameter α, β , γ , and δ are searched in the range $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. For our NCILFS algorithm, we determine the values of α , β , γ , and δ through grid search within the range $[10^{-3}, 10^{-2}, \dots, 10^2, 10^3]$. Finally, to further ensure the reliability and stability of the experimental results, we perform fivefold cross-validation for all methods. The experiments are conducted on a PC running Windows, with a CPU of 3.7 GHz and 32 GB of RAM. All algorithms are implemented in MATLAB.

To validate the effectiveness of NCILFS, we adopt five widely used multilabel learning metrics [11], [45] to evaluate model performance. These metrics are: *Average Precision (AP)*: which measures the average precision of predicted labels; *Hamming Loss (HL)*: which measures the proportion of incorrectly labeled samples; *Ranking Loss (RL)*: which evaluates the accuracy of the predicted label ranking; *Coverage (CV)*: which measures the average rank required to cover all correct labels; and *Micro-F1 (MiF)*: which assesses the balance between overall precision and recall of the model. Notably, for AP and MiF, higher values indicate better performance, whereas for HL, RL, and CV, lower values suggest better results.

Since the output of all algorithms is a ranking of feature importance, we select the top 20% of features from all features for performance testing in the experiments. In addition, we use the MLKNN ($K = 10$) classifier to evaluate model performance and quantitatively analyze the performance of each algorithm based on the AP, HL, RL, CV, and MiF metrics. For convenience in comparison, we denote the average value of a single metric across all benchmark datasets as ‘‘Average,’’ while the best result on a single dataset is marked in bold black.

B. Analysis of Experimental Results

Tables II–VI present the experimental results of NCILFS and the other nine comparison algorithms across five

TABLE II
EXPERIMENTAL RESULTS ON THE AP($\uparrow$)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12	Average
SSFS	0.7854 $\pm$ 0.0457	0.8374 $\pm$ 0.0038	0.7432 $\pm$ 0.0088	0.7407 $\pm$ 0.0097	0.6567 $\pm$ 0.0082	0.2922 $\pm$ 0.0015	0.6759
GRRO	0.8161 $\pm$ 0.0193	0.8549 $\pm$ 0.0044	0.7627 $\pm$ 0.0063	0.7363 $\pm$ 0.0063	0.6564 $\pm$ 0.0057	0.2931 $\pm$ 0.0070	0.6866
MDFS	0.7886 $\pm$ 0.0502	0.8503 $\pm$ 0.0050	0.7573 $\pm$ 0.0223	0.7474 $\pm$ 0.0077	0.6570 $\pm$ 0.0130	0.3016 $\pm$ 0.0139	0.6837
LRDG	0.8008 $\pm$ 0.0309	0.8549 $\pm$ 0.0041	0.7570 $\pm$ 0.0103	0.7555 $\pm$ 0.0048	0.6714 $\pm$ 0.0020	0.2921 $\pm$ 0.0047	0.6886
SRFS	0.8001 $\pm$ 0.0291	0.8504 $\pm$ 0.0042	0.7469 $\pm$ 0.0090	0.7495 $\pm$ 0.0049	0.6696 $\pm$ 0.0052	0.2941 $\pm$ 0.0044	0.6851
MSFS	0.8124 $\pm$ 0.0452	0.8481 $\pm$ 0.0018	0.7644 $\pm$ 0.0062	0.7514 $\pm$ 0.0103	0.6609 $\pm$ 0.0057	0.2892 $\pm$ 0.0137	0.6877
EF2FS	0.8078 $\pm$ 0.0469	0.8480 $\pm$ 0.0076	0.7628 $\pm$ 0.0091	0.7403 $\pm$ 0.0111	0.6613 $\pm$ 0.0083	0.2932 $\pm$ 0.0130	0.6856
I2VSLC	0.7874 $\pm$ 0.0144	0.8492 $\pm$ 0.0033	0.7546 $\pm$ 0.0163	0.7553 $\pm$ 0.0044	0.6755 $\pm$ 0.0063	0.3045 $\pm$ 0.0087	0.6878
GRAFS	0.8092 $\pm$ 0.0117	0.8535 $\pm$ 0.0040	0.7535 $\pm$ 0.0077	0.7576 $\pm$ 0.0066	0.6713 $\pm$ 0.0030	0.3050 $\pm$ 0.0044	0.6917
NCILFS	0.8197$\pm$0.0156	0.8551$\pm$0.0045	0.7695$\pm$0.0068	0.7635$\pm$0.0053	0.6787$\pm$0.0069	0.3117$\pm$0.0093	0.6997

TABLE III
EXPERIMENTAL RESULTS ON THE HL($\downarrow$)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12	Average
SSFS	0.2519 $\pm$ 0.0199	0.0965 $\pm$ 0.0014	0.2133 $\pm$ 0.0065	0.1683 $\pm$ 0.0048	0.0857 $\pm$ 0.0034	0.0174 $\pm$ 0.0001	0.1388
GRRO	0.1953 $\pm$ 0.0093	0.0880 $\pm$ 0.0015	0.1987 $\pm$ 0.0040	0.1726 $\pm$ 0.0025	0.0804 $\pm$ 0.0010	0.0173 $\pm$ 0.0003	0.1254
MDFS	0.2193 $\pm$ 0.0319	0.0884 $\pm$ 0.0022	0.2094 $\pm$ 0.0057	0.1630 $\pm$ 0.0040	0.0827 $\pm$ 0.0011	0.0199 $\pm$ 0.0008	0.1304
LRDG	0.2057 $\pm$ 0.0171	0.0886 $\pm$ 0.0023	0.2017 $\pm$ 0.0033	0.1623 $\pm$ 0.0022	0.0791 $\pm$ 0.0012	0.0174 $\pm$ 0.0002	0.1258
SRFS	0.2024 $\pm$ 0.0181	0.0905 $\pm$ 0.0025	0.2025 $\pm$ 0.0016	0.1651 $\pm$ 0.0031	0.0797 $\pm$ 0.0022	0.0174 $\pm$ 0.0002	0.1263
MSFS	0.2257 $\pm$ 0.0319	0.0898 $\pm$ 0.0024	0.2005 $\pm$ 0.0014	0.1616 $\pm$ 0.0039	0.0841 $\pm$ 0.0055	0.0199 $\pm$ 0.0005	0.1303
EF2FS	0.2438 $\pm$ 0.0318	0.0899 $\pm$ 0.0033	0.2047 $\pm$ 0.0035	0.1676 $\pm$ 0.0047	0.0832 $\pm$ 0.0020	0.0199 $\pm$ 0.0002	0.1348
I2VSLC	0.2052 $\pm$ 0.0043	0.0916 $\pm$ 0.0012	0.2067 $\pm$ 0.0130	0.1626 $\pm$ 0.0027	0.0799 $\pm$ 0.0015	0.0174 $\pm$ 0.0002	0.1272
GRAFS	0.1898 $\pm$ 0.0084	0.0893 $\pm$ 0.0017	0.2017 $\pm$ 0.0038	0.1622 $\pm$ 0.0031	0.0792 $\pm$ 0.0011	0.0173 $\pm$ 0.0001	0.1233
NCILFS	0.1804$\pm$0.0122	0.0877$\pm$0.0019	0.1977$\pm$0.0037	0.1590$\pm$0.0008	0.0778$\pm$0.0008	0.0173$\pm$0.0002	0.1200

TABLE IV
EXPERIMENTAL RESULTS ON THE RL($\downarrow$)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12	Average
SSFS	0.1815 $\pm$ 0.0321	0.0728 $\pm$ 0.0016	0.1855 $\pm$ 0.0056	0.1235 $\pm$ 0.0064	0.1625 $\pm$ 0.0108	0.1451 $\pm$ 0.0026	0.1452
GRRO	0.1552 $\pm$ 0.0210	0.0621 $\pm$ 0.0020	0.1717 $\pm$ 0.0054	0.1272 $\pm$ 0.0038	0.1618 $\pm$ 0.0039	0.1418 $\pm$ 0.0029	0.1366
MDFS	0.1773 $\pm$ 0.0563	0.0653 $\pm$ 0.0026	0.1742 $\pm$ 0.0144	0.1232 $\pm$ 0.0048	0.1626 $\pm$ 0.0093	0.1479 $\pm$ 0.0062	0.1418
LRDG	0.1658 $\pm$ 0.0320	0.0623 $\pm$ 0.0026	0.1733 $\pm$ 0.0072	0.1168 $\pm$ 0.0022	0.1537 $\pm$ 0.0013	0.1447 $\pm$ 0.0037	0.1361
SRFS	0.1671 $\pm$ 0.0237	0.0638 $\pm$ 0.0026	0.1789 $\pm$ 0.0071	0.1195 $\pm$ 0.0021	0.1557 $\pm$ 0.0022	0.1446 $\pm$ 0.0032	0.1383
MSFS	0.1705 $\pm$ 0.0490	0.0670 $\pm$ 0.0016	0.1684 $\pm$ 0.0058	0.1180 $\pm$ 0.0063	0.1566 $\pm$ 0.0047	0.1547 $\pm$ 0.0078	0.1392
EF2FS	0.1624 $\pm$ 0.0299	0.0650 $\pm$ 0.0025	0.1707 $\pm$ 0.0037	0.1251 $\pm$ 0.0055	0.1647 $\pm$ 0.0108	0.1493 $\pm$ 0.0073	0.1395
I2VSLC	0.1784 $\pm$ 0.0109	0.0637 $\pm$ 0.0016	0.1733 $\pm$ 0.0132	0.1164 $\pm$ 0.0027	0.1522 $\pm$ 0.0059	0.1424 $\pm$ 0.0031	0.1377
GRAFS	0.1552 $\pm$ 0.0072	0.0630 $\pm$ 0.0019	0.1765 $\pm$ 0.0060	0.1158 $\pm$ 0.0049	0.1532 $\pm$ 0.0051	0.1385 $\pm$ 0.0012	0.1337
NCILFS	0.1442$\pm$0.0176	0.0618$\pm$0.0024	0.1620$\pm$0.0056	0.1124$\pm$0.0030	0.1498$\pm$0.0048	0.1347$\pm$0.0034	0.1275

TABLE V
EXPERIMENTAL RESULTS ON THE CV($\downarrow$)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12	Average
SSFS	2.0221 $\pm$ 0.1973	12.0670 $\pm$ 0.1104	6.6754 $\pm$ 0.0759	19.2402 $\pm$ 0.6165	6.5209 $\pm$ 0.2371	111.9968 $\pm$ 1.2662	26.4204
GRRO	1.7945 $\pm$ 0.1001	11.3502 $\pm$ 0.1831	6.4010 $\pm$ 0.2195	19.2068 $\pm$ 0.2356	6.4507 $\pm$ 0.0674	110.2908 $\pm$ 2.0009	25.9157
MDFS	1.9700 $\pm$ 0.3960	11.4337 $\pm$ 0.1990	6.6044 $\pm$ 0.2693	19.3050 $\pm$ 0.5265	6.5378 $\pm$ 0.3513	124.0339 $\pm$ 3.6054	28.3141
LRDG	1.7828 $\pm$ 0.1344	11.3236 $\pm$ 0.3174	6.3999 $\pm$ 0.1216	18.8229 $\pm$ 0.2074	6.2005 $\pm$ 0.0859	112.8451 $\pm$ 2.8906	26.2291
SRFS	1.7926 $\pm$ 0.0770	11.4418 $\pm$ 0.2247	6.4380 $\pm$ 0.1125	19.0056 $\pm$ 0.1518	6.2657 $\pm$ 0.0866	111.9207 $\pm$ 1.5449	26.1441
MSFS	1.9536 $\pm$ 0.3520	11.4524 $\pm$ 0.2779	6.4038 $\pm$ 0.0830	18.9122 $\pm$ 0.3861	6.4896 $\pm$ 0.1093	126.9432 $\pm$ 7.7324	28.6925
EF2FS	1.9631 $\pm$ 0.1621	11.3719 $\pm$ 0.2235	6.3860 $\pm$ 0.1357	19.2723 $\pm$ 0.4789	6.6499 $\pm$ 0.3465	121.7045 $\pm$ 4.3177	27.8913
I2VSLC	1.8957 $\pm$ 0.1205	11.3427 $\pm$ 0.1599	6.3913 $\pm$ 0.1713	18.8337 $\pm$ 0.2680	6.1475 $\pm$ 0.1712	110.7984 $\pm$ 1.7147	25.9016
GRAFS	1.7324 $\pm$ 0.0628	11.4007 $\pm$ 0.2031	6.4489 $\pm$ 0.1732	18.7414 $\pm$ 0.4346	6.1787 $\pm$ 0.2092	108.9515 $\pm$ 1.0125	25.5756
NCILFS	1.7042$\pm$0.0994	11.2600$\pm$0.1860	6.2317$\pm$0.0828	18.6069$\pm$0.3440	6.1150$\pm$0.1450	107.4893$\pm$2.7335	25.2345

evaluation metrics. Each table shows the average and standard deviation values of each algorithm on different datasets. From Tables II–VI, we can observe that the NCILFS algorithm achieves the best results in AP, HL, RL, and CV metrics across all six datasets. In the MiF metric, NCILFS gets the best results on five datasets, with only one dataset showing slightly inferior performance compared to other algorithms. From the average results across all metrics, the performance

of NCILFS is significantly better than that of the other comparison algorithms. Overall, the excellent performance of NCILFS across multiple metrics reflects its strong capability in mining the consistency and complementary information in multiview multilabel data.

In addition, we conduct feature addition experiments to analyze the performance trends of different algorithms across five evaluation metrics. Due to space limitations, we present

TABLE VI
EXPERIMENTAL RESULTS ON THE MiF(†)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12	Average
SSFS	0.5579±0.0444	0.7282±0.0041	0.5851±0.0127	0.6110±0.0143	0.4873±0.0126	0.0437±0.0095	0.5022
GRRO	0.6513±0.0177	0.7596±0.0044	0.6289±0.0077	0.6017±0.0055	0.5109±0.0077	0.0482±0.0100	0.5334
MDFS	0.6013±0.0731	0.7514±0.0055	0.5962±0.0165	0.6190±0.0112	0.5073±0.0115	0.0995±0.0202	0.5291
LRDG	0.6459±0.0240	0.7585±0.0063	0.6253±0.0078	0.6331±0.0081	0.5270±0.0095	0.0468±0.0123	0.5394
SRFS	0.6550±0.0239	0.7535±0.0084	0.6181±0.0036	0.6306±0.0034	0.5213±0.0160	0.0453±0.0077	0.5373
MSFS	0.6133±0.0638	0.7451±0.0051	0.6172±0.0054	0.6174±0.0107	0.4986±0.0291	0.0648±0.0145	0.5261
EF2FS	0.5747±0.0682	0.7483±0.0086	0.6105±0.0139	0.6145±0.0137	0.4992±0.0145	0.0681±0.0132	0.5192
I2VSLC	0.6501±0.0144	0.7471±0.0025	0.6141±0.0240	0.6341±0.0081	0.5218±0.0115	0.0663±0.0061	0.5389
GRAFS	0.6730±0.0210	0.7556±0.0038	0.6244±0.0079	0.6362±0.0051	0.5246±0.0081	0.0605±0.0032	0.5457
NCILFS	0.6841±0.0263	0.7614±0.0061	0.6309±0.0110	0.6390±0.0067	0.5307±0.0022	0.0579±0.0115	0.5507

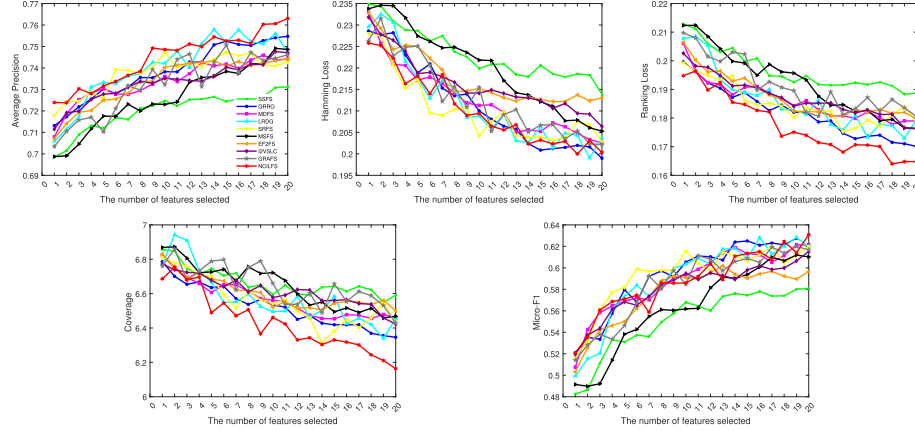


Fig. 2. Performance of different algorithms on the Yeast dataset.

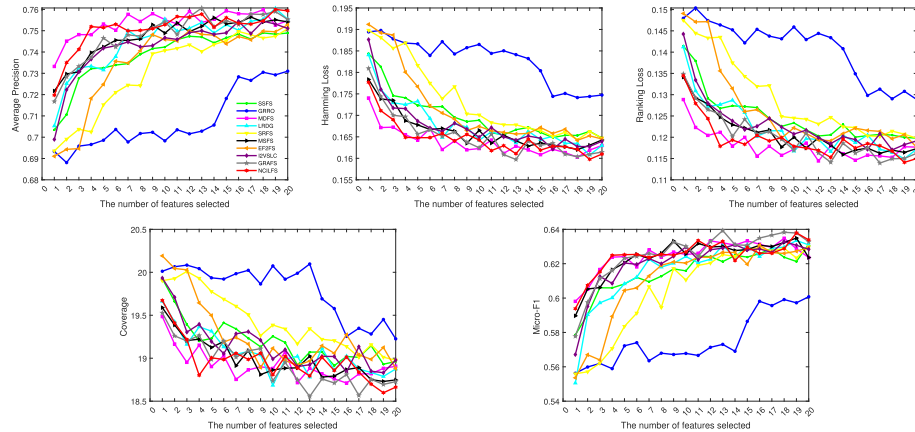


Fig. 3. Performance of different algorithms on the Mirflickr dataset.

detailed results for the Yeast and Mirflickr datasets only in Figs. 2 and 3. In the figures, the horizontal axis represents the number of selected features, ranging from 1% to 20% of all features, with a step size of 1%; the performance values of the evaluation metrics are shown on the vertical axis. As shown in the figures, as the number of selected features increases, the performance of all algorithms on the five evaluation metrics exhibits a trend of “initial increase followed by stabilization.” Further analysis of Figs. 2 and 3 reveal that, on the Yeast dataset, NCILFS has the best overall performance curves across all metrics. On the Mirflickr dataset, the performance of NCILFS is comparable to MDFS, MSFS, and GRAFS in the initial stage, but as the number of features increases,

the performance of NCILFS gradually surpasses the other algorithms. In summary, the proposed algorithm demonstrates better and more stable performance.

C. Statistical Analysis

To further evaluate the performance differences between different methods, we apply the Friedman test [46] and the Bonferroni–Dunn test [47]. These two statistical tests are commonly used for multialgorithm performance comparisons and can effectively measure the relative performance of different algorithms across multiple datasets. Let the number of datasets be represented by N_d and the number of algorithms by A .

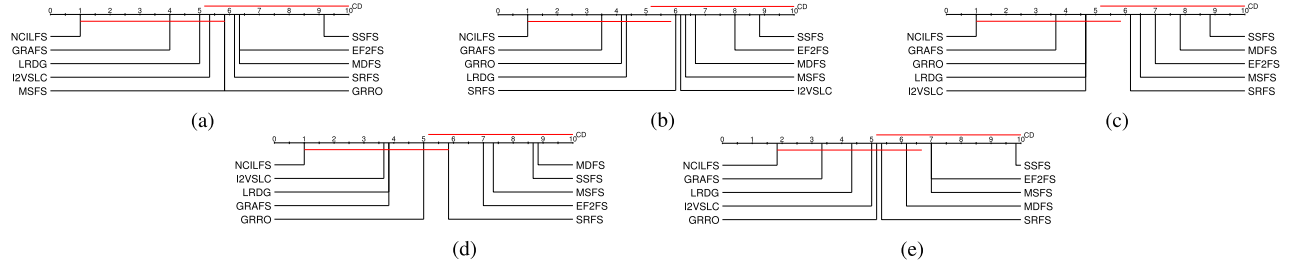


Fig. 4. Comparison of NCILFS against other comparison algorithms with the Bonferroni-Dunn test. (a) AP. (b) HL. (c) RL. (d) CV. (e) MiF.

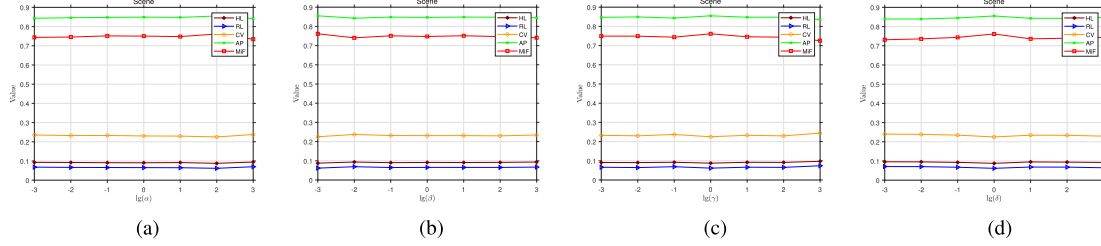


Fig. 5. Parameter sensitivity of the Scene dataset. (a) α . (b) β . (c) γ . (d) δ .

TABLE VII

FRIEDMAN STATISTICS F_F ($A = 10$, $N_d = 6$) AND CRITICAL VALUES ON FIVE METRICS

Evaluation Metric	F_F	Critical Value ($\alpha = 0.05$)
Average Precision	4.3279	2.096
Hamming Loss	6.7857	
Ranking Loss	6.2842	
Coverage	10.7309	
Micro-F1	5.6071	

Under the null hypothesis (i.e., no significant performance differences among all algorithms), the Friedman statistic can be expressed as

$$\chi_F^2 = \frac{12N_d}{A(A+1)} \left(\sum_{i=1}^A \left(\frac{\sum_{j=1}^{N_d} R_i^j}{N_d} \right)^2 - \frac{A(A+1)^2}{4} \right) \quad (35)$$

$$F_F = \frac{(N_d - 1)\chi_F^2}{N_d(A - 1) - \chi_F^2} \quad (36)$$

where R_i^j represents the rank of the i th algorithm on the j th dataset. The F_F and the corresponding critical value for each evaluation metric are summarized in Table VII. At a significance level of $\alpha = 0.05$, the critical value is 2.096. As shown in the table, the F_F values for each evaluation metric are greater than the critical value, indicating a clear rejection of the null hypothesis.

Next, we further analyze the results of each algorithm using the Bonferroni-Dunn test. First, the critical difference (CD) is calculated using the following formula:

$$CD_\alpha = q_\alpha \sqrt{\frac{A(A+1)}{6N_d}} \quad (37)$$

where q_α is the constant associated with the significance level α , and at $\alpha = 0.05$, its value is 2.773. Based on this formula, we calculate the critical difference $CD = 4.8472$. Fig. 4 presents the Bonferroni-Dunn test results for the five metrics, with the red line indicating the range of CD. If an algorithm falls outside the red line, it is considered to have a significant performance difference compared to the proposed algorithm. From the analysis of Fig. 4, it is clear that NCILFS significantly outperforms SSFS, MDIFS, SRFS, MSFS, and EF2FS on the AP, HL, RL, and CV metrics. It also shows superior performance on MiF compared to SSFS, MSFS, and EF2FS. Furthermore, NCILFS achieves the highest average ranking across all five evaluation metrics, indicating that its overall performance surpasses that of the other algorithms. In conclusion, NCILFS demonstrates the best performance among all methods, showcasing its superior capabilities.

D. Parameter Sensitivity Analysis

There are four parameters in our objective function. To study the impact of each parameter on the feature selection performance, we conduct a parameter sensitivity analysis. Specifically, when analyzing a particular parameter, we fix the other parameters at their optimal values to ensure the reliability of the analysis results. Since the evaluation metrics for different datasets have varying scales, to visually display the trends of all metrics in the same plot, we scale the CV metric for the Scene and Mirflickr datasets to 0.02 times their original value, and for the Yeast dataset, we scale the CV metric to 0.1 times its original value. By analyzing the parameter sensitivity plots (as shown in Figs. 5–7), the following conclusions can be drawn: the values of all metrics for the Scene and Yeast datasets show little variation with changes in the parameters, indicating high stability. Experimental results show that optimal performance is achieved when α is set to a relatively large value, while β , γ , and δ are assigned moderate values. In the Mirflickr dataset, certain metrics show some

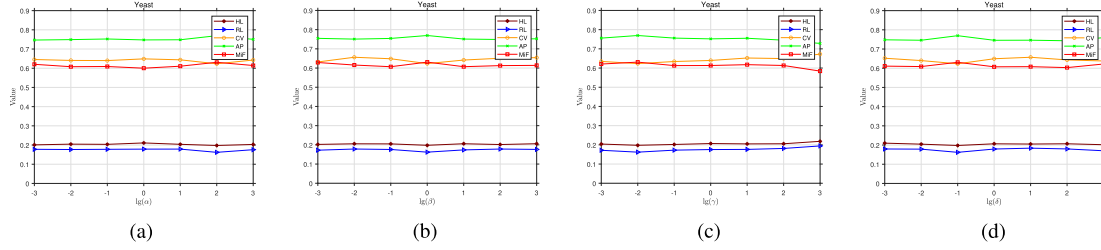
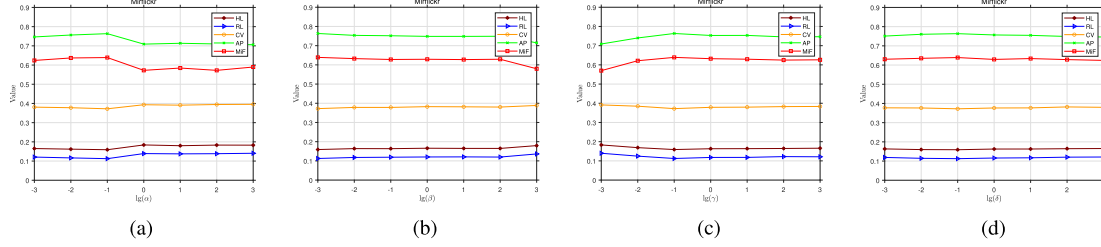
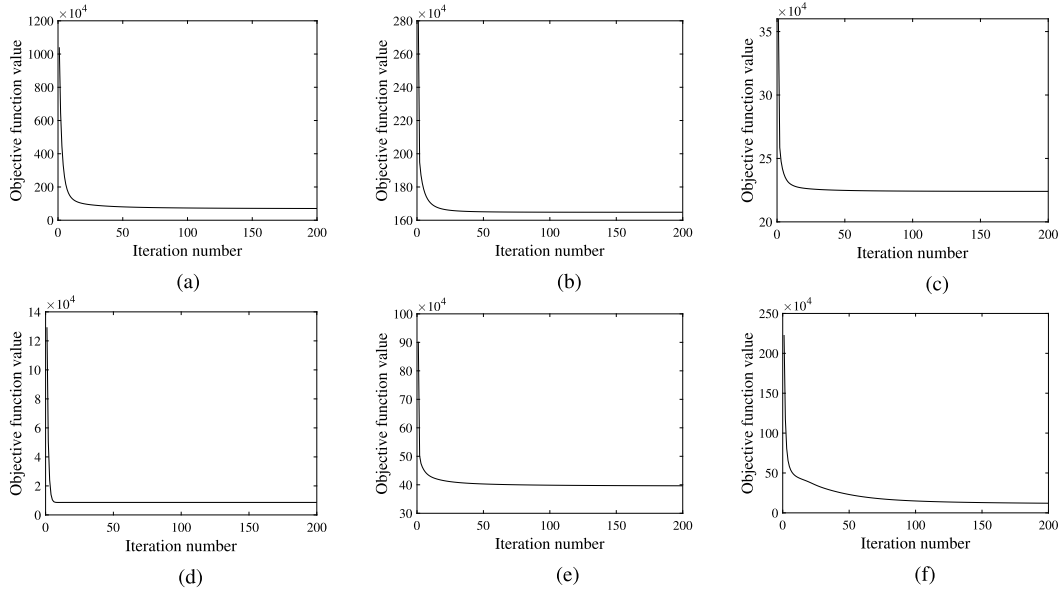
Fig. 6. Parameter sensitivity of the Yeast dataset. (a) α . (b) β . (c) γ . (d) δ .Fig. 7. Parameter sensitivity of the Mirflickr dataset. (a) α . (b) β . (c) γ . (d) δ .

Fig. 8. Convergence curves. (a) Emotions. (b) Scene. (c) Yeast. (d) Mirflickr. (e) Voc07. (f) Iaprtc12.

variation within a small parameter range, but the changes do not exceed 10%. In this case, the best performance is obtained when α is set to a relatively small value and the other parameters remain moderate. Overall, setting α, β, γ , and δ to moderate values generally yields optimal or near-optimal results across most cases, while in certain situations, assigning a marginal value to α can further improve performance.

E. Convergence and Running Time Analysis

In Section IV, we prove theoretically that the proposed algorithm (NCILFS) converges. To further verify its practical convergence efficiency, we conduct experiments on six datasets and plot the curve of the objective function value against the number of iterations (as shown in Fig. 8). From the figure, it can be seen that the value of the objective function decreases

rapidly during the iterations, indicating that the algorithm has good convergence speed. Furthermore, the values of the objective function for most datasets stabilize within 40 iterations, which further verifies the high convergence efficiency of NCILFS.

In addition, we present the running times of the algorithms on six datasets in Table VIII to provide a visual comparison of the computational efficiency of each method. SSFS, SRFS, MSFS, EF2FS, I2VSLC, GRAFS, and NCILFS are typical embedded methods, so their running times are similar. GRRO requires the calculation of a large amount of mutual information, which results in its running time being significantly slower than the other algorithms. LRDG incorporates dynamic graph constraints, leading to a higher computational cost. NCILFS runs slower on Iaprtc12, which may be due to the larger number of labels in this dataset, with the learning of the

TABLE VIII
RUNNING TIME (IN SECOND)

Dataset	Emotions	Scene	Yeast	Mirflickr	Voc07	Iaprtc12
SSFS	0.03	0.34	0.08	1.52	1.41	0.80
GRRO	0.36	96.17	2.03	100.94	99.43	513.72
MDFS	0.17	24.37	3.73	21.14	11.59	56.15
LRDG	0.17	25.61	3.98	25.09	20.25	47.31
SRFS	0.01	0.53	0.03	0.39	0.53	3.18
MSFS	0.02	1.53	0.20	1.62	2.23	2.92
EF2FS	0.04	5.96	0.53	3.33	2.02	15.41
I2VSLC	0.03	2.44	0.19	1.81	0.83	15.24
GRAFS	0.05	5.76	0.71	5.56	5.49	27.54
NCILFS	0.13	2.21	0.27	1.21	1.58	115.87

three types of labels consuming additional time. Overall, our method is able to complete the feature selection task efficiently within a reasonable amount of time.

VI. CONCLUSION

This article proposed a feature selection method for multi-view multilabel data. Specifically, we first learn a commonality label matrix by extracting shared information from the feature spaces of each view. Then, considering the differences between views and the influence of label noise, we introduce individuality label matrices and view noise label matrices, and constrain the learning of these matrices using manifold regularization and the l_1 -norm. Finally, to further reduce the noise contribution and enhance the role of important views, we determine the weight of each view based on fuzzy mutual information and design an efficient optimization method to ensure that the objective function converges quickly during the iteration process. The effectiveness and superiority of NCILFS are demonstrated by a series of experimental results. In future work, we will simultaneously consider the effects of sample noise and label noise and explore a more robust feature selection method to better adapt to complex multiview multilabel data scenarios.

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